ECA5403-ELECTROMAGNETIC FDIELDS AND WAVES

CO-1

EXPERIMENT 3

AIM

To determine and plot the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB.

PROGRAM

Test case (i) Determination and plotting of force between two electric charges with varying distance

```
clc;

clear;

q1 = input('Enter the charge q1=');

q2 = input('Enter the charge q2=');

epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);

r = linspace(0.1,10,100);

F = k*q1*q2./(r.^2);

disp('Force at r = 1 m is:');

disp(k*q1*q2/(1^2));

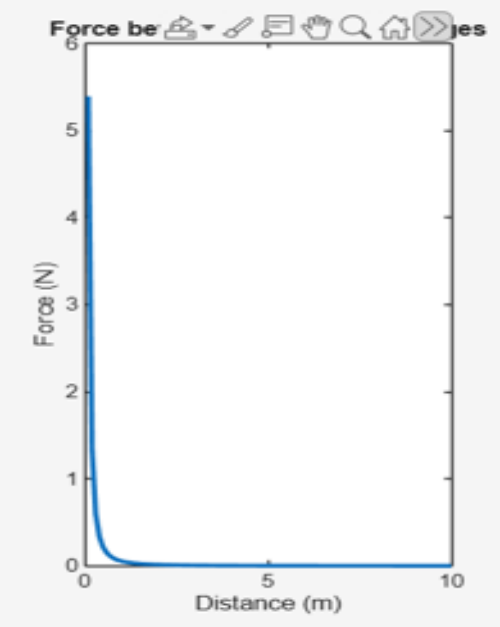
plot(r,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force (N)');

title('Force between two electric charges');
```

OUTPUT



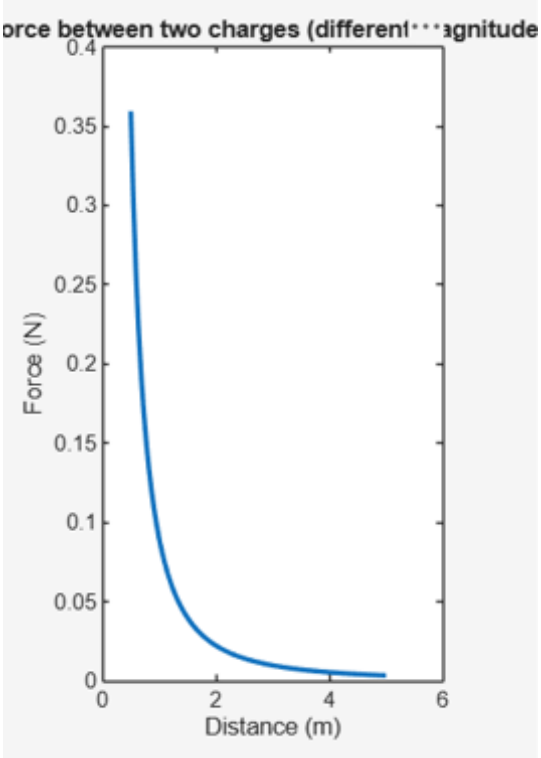
Test case (ii) Determination and plotting of force between two electric charges with different

magnitudes

Code:

```
clc;
q1 = 5e-6;
q2 = 2e-6;
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
r = linspace(0.5,5,100);
F = k*q1*q2./(r.^2);
plot(r,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force (N)');
title('Force between two charges (different magnitudes)');
```

OUTPUT



Test case (iii) Determination and plotting of force per unit length between two parallel

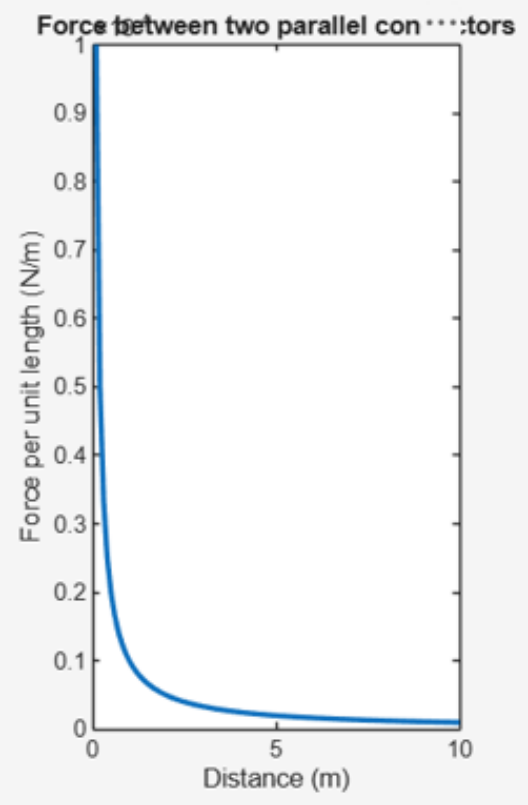
conductors with varying distance

Code:

```
clc;
mu0 = 4*pi*1e-7;
I1 = 5;
I2 = 10;
d = linspace(0.1,10,100);
F = (mu0*I1*I2)./(2*pi*d);
plot(d,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force per unit length (N/m)');
title('Force between two parallel
```

conductors');

OUTPUT

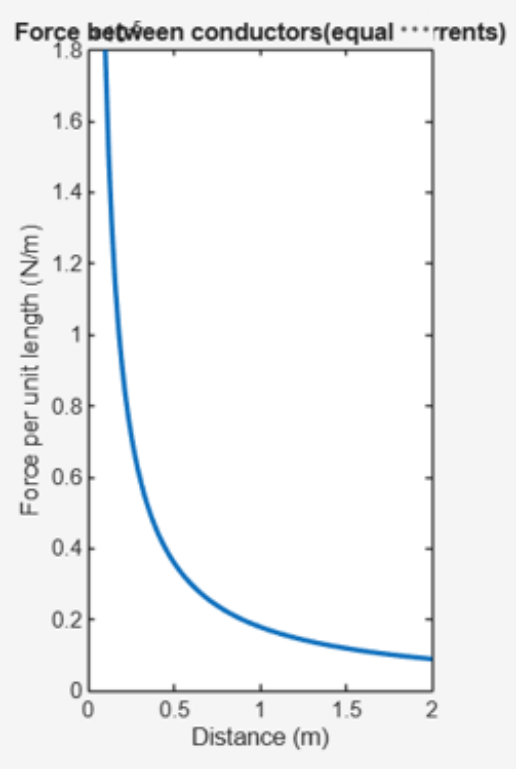


Test case (iv) Determination and plotting of force between two parallel conductors carrying equal currents

```
clc;
mu0 = 4*pi*1e-7;
I1 = 3;
I2 = 3;
d = linspace(0.1,2,100);
F = (mu0*I1*I2)./(2*pi*d);
plot(d,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force per unit length (N/m)');
```

title('Force between conductors

(equal currents)');



RESULT

Thus the program was verified for the force of attraction or repulsion acting along a straight linebetween two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them using SCILAB was successfully.

RUBRICS

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks given	Marks Obtained
Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifies most quantities and models correctly.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities.	25	2.5	
Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel-current force calculations correctly with units.	Calculations are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25	2.5	
Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence.	Trend and plots are correct with minor omissions.	Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship.	30	3	
Explain and report results	Explain and report results	Explains attraction/repulsion and reports assumptions, units and conclusions clearly.	Conclusion is correct with minor documentation gaps.	Conclusion is incomplete or partly inaccurate.	Conclusion is missing or incorrect.	20	2	
TOTAL						100	10	